Engine Model(s): PT6A

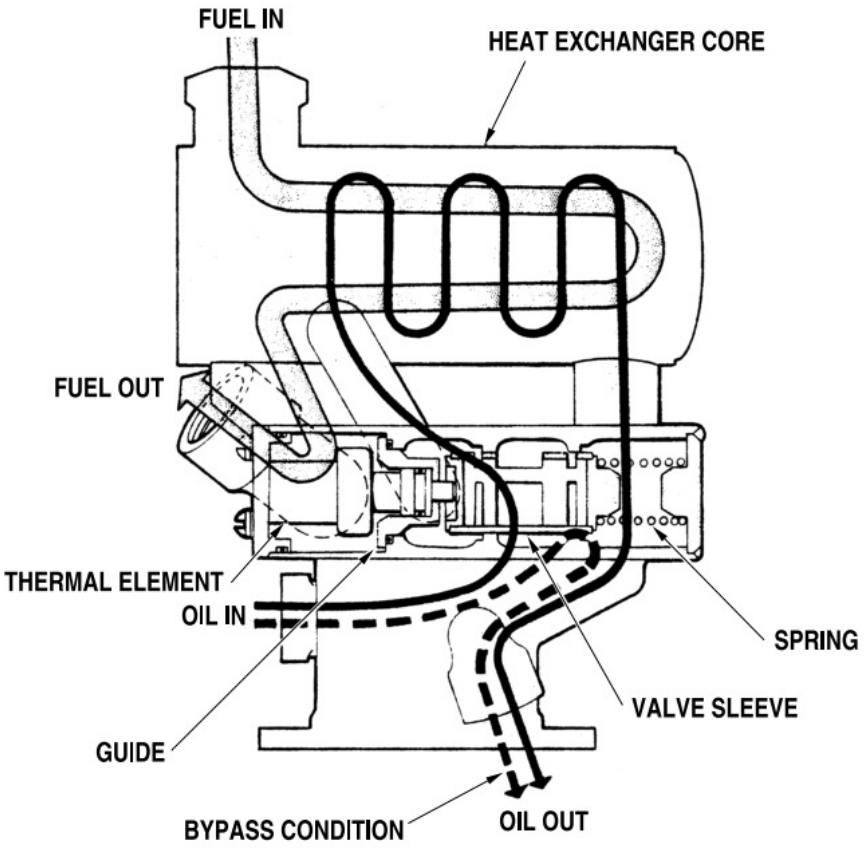
73-10-01

OIL-TO-FUEL HEATER - DESCRIPTION AND OPERATION

1. Description (Ref. Fig. 1)

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Figure 1 Oil-to-Fuel Heater Schematic



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The oil-to-fuel heater, mounted at the top of the accessory gearbox at the rear of the engine and bolted to Flange G, is essentially a heat exchanger which utilizes heat from the engine lubricating oil system to preheat the fuel in the fuel system. The heater incorporates in honeycomb two-pass oil circuit and a two-pass fuel circuit. A fuel temperature-sensing oil bypass valve regulates the fuel temperature by either allowing oil to flow through the heater circuit, or to bypass it. Oil flow from the outlet passes through the oil return check valve and back to tank.

2. Operation

A minimum pressurizing valve (Ref. 79-20-03), fitted in a port at the 2 o'clock position on the compressor inlet case is connected by rigid tubing to the inlet side of the oil-to-fuel heater oil circuit. The valve closes at approximately 40 psig to prevent oil flow to the heater when the engine is shut down in flight when the compressor is windmilling. A check valve in the oil-to-fuel heater oil-return-to-tank prevents any oil from the airframe cooler being applied to the heater when the engine is shut down.

The temperature-sensing oil bypass (thermal element) valve consists of a highly expansive material sealed in a metallic chamber. The expansion force is transmitted through a diaphragm and plunger to a piston. Since the element only exerts an expansive force, it is counterbalanced by a return spring which provides a contracting force during decreases in temperature. The element senses the temperature of the outlet fuel and at temperatures above 21°C (70°F), starts to close the core valve and simultaneously open the bypass valve. At 32°C (90°F), the core valve is completely closed and oil bypasses the heater core.

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